

LABORATORY 4**COMMON SOURCE SINGLE STAGE AMPLIFIER****OBJECTIVES**

No.	Objectives	Requirements
1	Understand the operation of the common-source (CS) amplifier and the effect of biasing on gain and swing	Set device sizes to achieve proper biasing and verify gain, output swing, and input swing
2	Analyze the relationship between gain, output swing, and DC operating point.	Verify the amplifiers with specific bias condition
3	Learn the design flow of an amplifier	- Design amplifiers to meet given specifications. - Verify gain and bandwidth (BW), and simulate the sine-wave response

LAB 4 INFORMATION

- Consider *NMOS nmos1v_lvt device*, *PMOS pmos1v_lvt device* in *gpd45nm* and power supply $v_{dd} = 1.0V$
- Note:** All figures in the document are for reference only.



EXPERIMENT 1

1.1. Objective: Get knowledge about how the bias conditions affect to an amplifier.

1.2. Requirements: Simulate dc analysis with ADE-Explorer environment.

1.3. Instruction: Assemble the circuit as shown in **Figure 1**, setting parameters for power supplies (using *vdc* source), with $L_n = L_p = 45\text{nm}$, $W_{n,\text{unit}} = W_{p,\text{unit}} = 120\text{nm}$, $v_{dd} = 1.0\text{V}$, $V_{bias} = 0.6\text{V}$, $C_{load} = 3\text{fF}$.

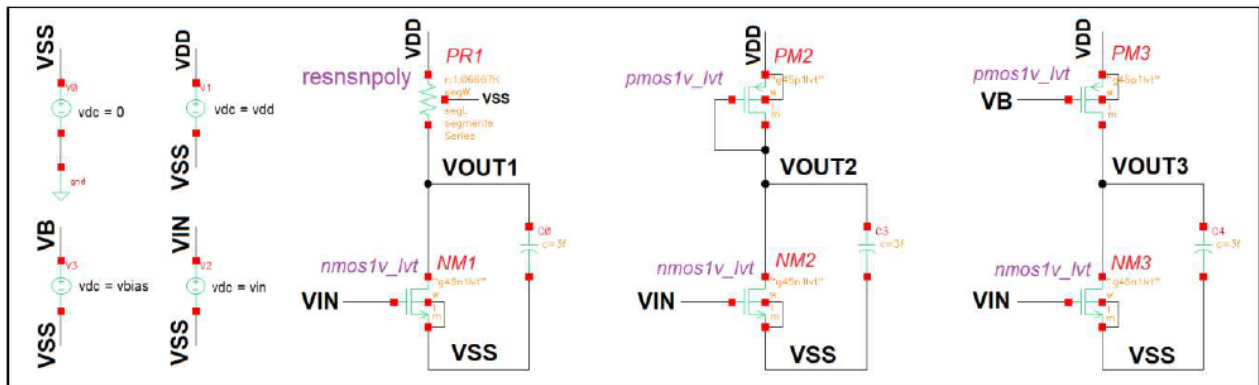


Figure 1. Common source stage schematic for DC analysis

1.4. Analyze:➤ **1.4.1. Testcase:**

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT
Temp	25°C
V_{DD}	1 V
V_{in}	$0 \rightarrow V_{DD}$
V_{bias}	$V_{bias} = 0.6\text{V}$
Observe	IN-OUT characteristic curve

- Set up device sizes in Table 1 for each one:

Device name	Size
NM1	$L_n = 45\text{nm}$, $W_n = 120\text{nm}$, $fingers = 1$, $multipliers = 1$

PR1	Segment Width = 1u, Segment Length = 8u Segment = 20 , Segment Connection = Series
NM2	$L_n = 45nm, W_n = 120nm,$ $fingers = 1, multipliers = 1$
PM2	$L_p = 45nm, W_p = 120nm,$ $fingers = 1, multipliers = 2$
NM3	$L_n = 45nm, W_n = 120nm,$ $fingers = 1, multipliers = 1$
PM3	$L_p = 45nm, W_p = 120nm,$ $fingers = 1, multipliers = 32$

- Using dc analysis, sweeping variable v_{in} from 0V to v_{dd} , step = 0.01V as shown in **Figure 2**. Plot IN-OUT characteristic curve as **Figure 4**.

Analysis: ☐ tran ☒ dc ☐ ac ☐ noise
☐ xf ☐ sens ☐ dcmatch ☐ acmatch
☐ stb ☐ pz ☐ lf ☐ sp
☐ envlp ☐ pss ☐ pac ☐ pstb
☐ pnoise ☐ pxf ☐ psp ☐ qpss
☐ qpac ☐ qpnoise ☐ qpxf ☐ qpssp
☐ hb ☐ hbac ☐ hbstb ☐ hbnoise
☐ hbasp ☐ hbxf ☐ ofa

DC Analysis

Save DC Operating Point ☐
Hysteresis Sweep ☐

Sweep Variable

☐ Temperature
☒ Design Variable Variable Name:
☐ Component Parameter
☐ Model Parameter

Sweep Range

☒ Start-Stop Start: Stop:
☐ Center-Span

Sweep Type

☒ Linear ☐ Step Size
☐ Number of Steps

Figure 2. Setup for DC analysis

➤ 1.4.2. Analysis:

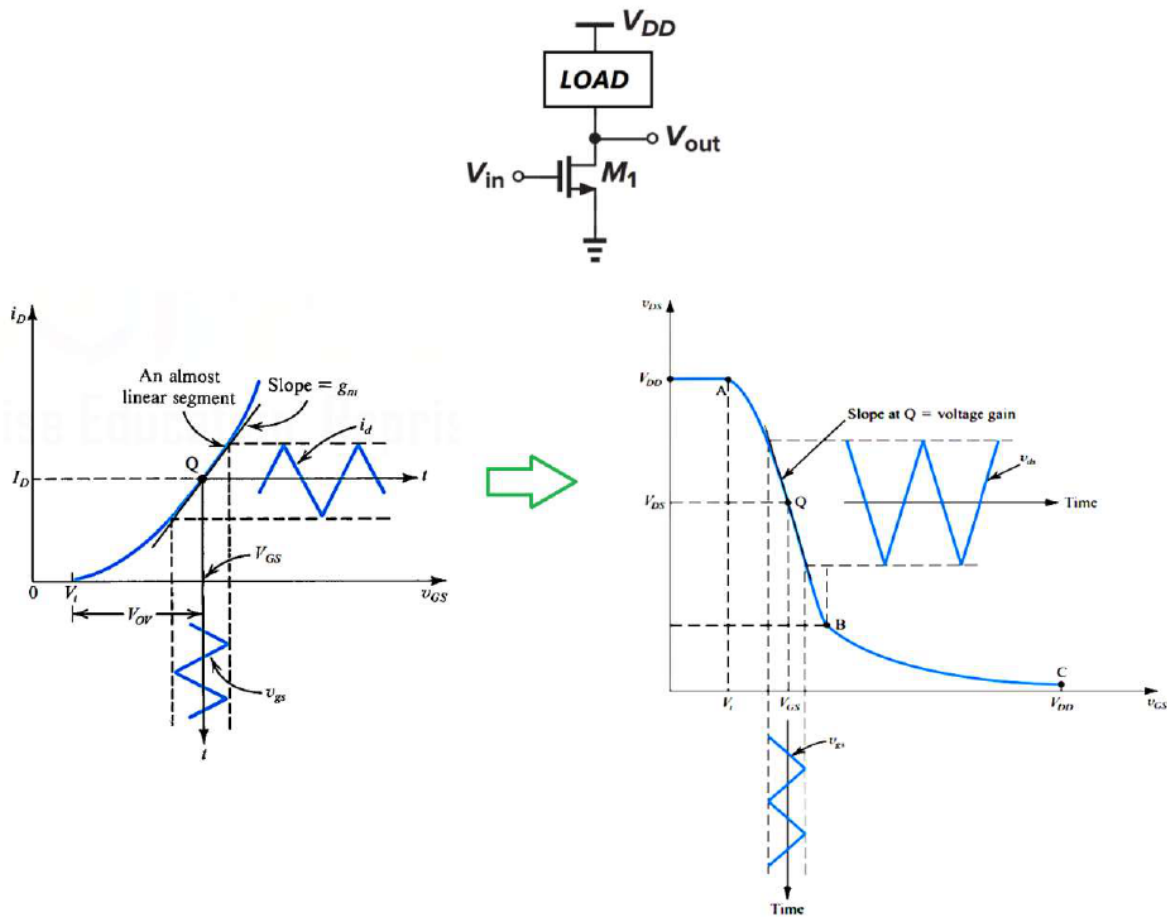


Figure 3. Analyze the IN-OUT response of an amplifier with CS topology

- The transistor M1 converts variations in the input voltage V_{in} into variations in the drain current I_D (transconductance behavior - g_m).
- The load then converts the change in drain current ΔI_D into a change in output voltage ΔV_{OUT} .

$$\Delta V_{IN} \xrightarrow{g_m} \Delta I_D \xrightarrow{Z_{load}} \Delta V_{OUT}$$

- For proper operation, NMOS M1 must be biased in the active region to ensure a well-defined slope (Gain). This process is called biasing.
- In Experiment 1, students will study the fundamental relationship between:
 - Voltage gain
 - Output voltage swing (Headroom)
 - Input voltage swing
 - Output DC operating point (bias level)

- Several equations are used to determine the gain, output swing, input swing, and output DC operating point.

- Gain:

$$Gain = \frac{\Delta V_{OUT}}{\Delta V_{IN}} = -g_m Z_{load}$$

- Output voltage max-swing (Headroom):

$$V_{OUT}(B) - V_{OUT}(A),$$

which A, B are the 2 points in **Figure 3**

- Input voltage max-swing:

$$V_{IN}(B) - V_{IN}(A),$$

which A, B are the 2 points in **Figure 3**

- Output DC operating point (bias level) at Q in **Figure 3**

$$V_{OUT}(Q) = V_{DD} - I_D Z_{load} = V_{DD} - I_D \cdot \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{IN}(Q) - V_{TH})^2$$

- For maximum swing and improved gain, the DC operating point Q is typically set at the midpoint of the maximum output and input voltage swings.

➤ 1.4.3. Waveform:

- The Figure 4 depicts the IN-OUT characteristic curve.

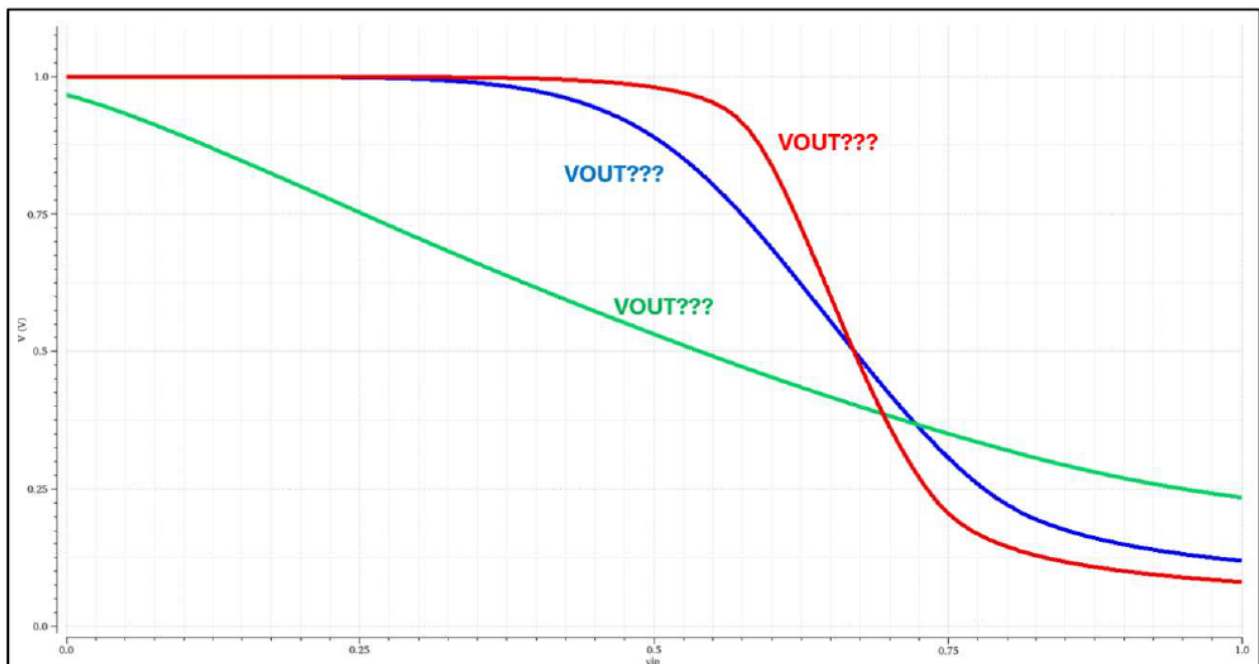


Figure 4. IN-OUT response of an amplifier with CS topology

- Determine which curve corresponds to each topology (red, blue and green)

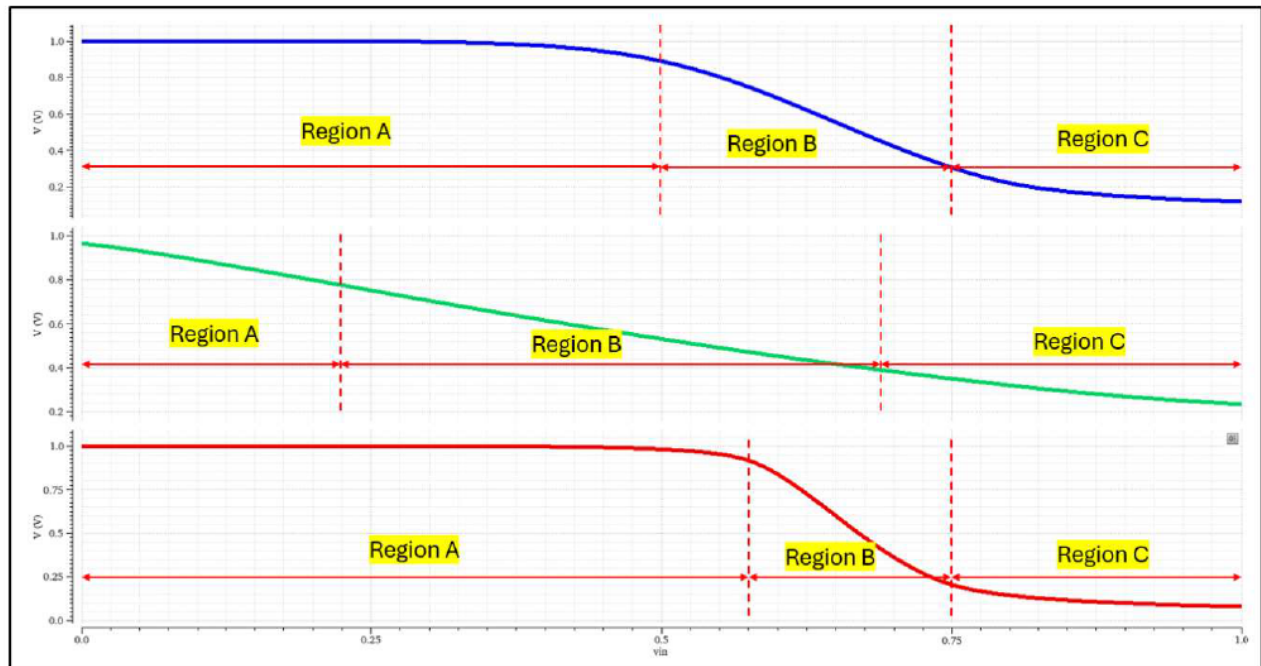


Figure 5. IN-OUT response of an amplifier with CS topology

- Determine the operating region of transistor M1 at Region A, Region B and Region C in Figure 5.

Curve	Topology	Region	From To	Region of M1
Red	Resistive/ Diode connected/ Current source	Region A	From ... mV to ... mV	Cut off/Triode/Saturation
		Region B	From ... mV to ... mV	Cut off/Triode/Saturation
		Region C	From ... mV to ... mV	Cut off/Triode/Saturation
Blue	Resistive/ Diode connected/ Current source	Region A	From ... mV to ... mV	Cut off/Triode/Saturation
		Region B	From ... mV to ... mV	Cut off/Triode/Saturation
		Region C	From ... mV to ... mV	Cut off/Triode/Saturation
Green	Resistive/ Diode connected/ Current source	Region A	From ... mV to ... mV	Cut off/Triode/Saturation
		Region B	From ... mV to ... mV	Cut off/Triode/Saturation
		Region C	From ... mV to ... mV	Cut off/Triode/Saturation

➤ 1.4.4. Extract from waveform:

- Using DC analysis, sweeping variable v_{in} from 0V to v_{dd} , step = 0.01V as shown in **Figure 2**. Plot IN-OUT characteristic curve as **Figure 4**.
- Method to calculate DC Gain, input max swing and output max swing as shown in **Figure 6**.

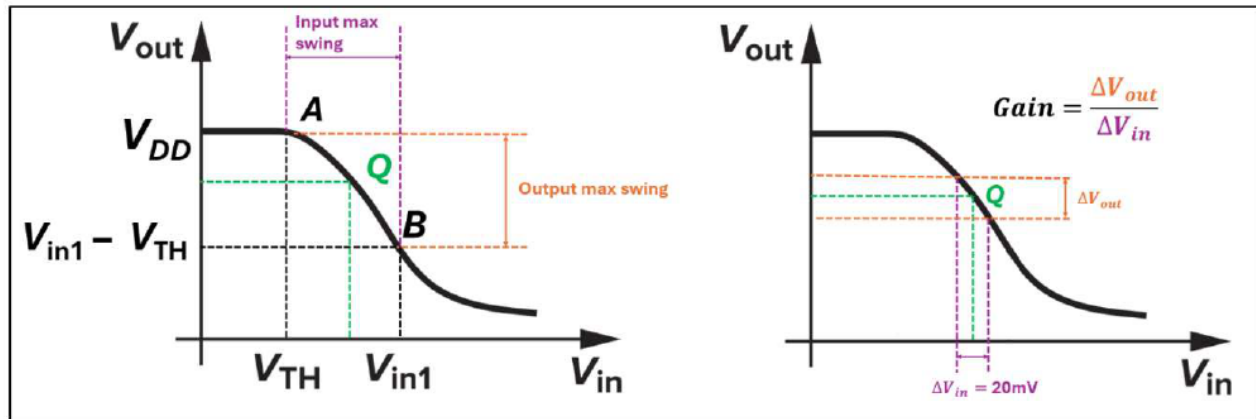


Figure 6. Method calculating Gain, Input max-swing and Output max-swing

Topology	Optimize Q	Gain (dB) at Q with $\Delta V_{IN} = 20mV$	Max swing input	Max swing output
Resistive	$V_{IN} = \dots$ $V_{OUT} = \dots$			
Diode connected	$V_{IN} = \dots$ $V_{OUT} = \dots$			
Current source	$V_{IN} = \dots$ $V_{OUT} = \dots$			

- Compare the benefits and drawbacks of 3 topologies.**

➤ 1.4.5. Explanation (if applicable):

EXPERIMENT 2

2.1. Objective: Verify size devices with target bias condition.

2.2. Requirements: Simulate dc analysis with ADE-Explorer environment.

2.3. Instruction: Assemble the circuit as shown in **Figure 1**, setting parameters for power supplies (using **vdc** source), with $L_n = L_p = 45\text{nm}$, $W_{n,\text{unit}} = W_{p,\text{unit}} = 120\text{nm}$, $v_{dd} = 1.0\text{V}$, $V_{in,DC} = 0.7\text{V}$, $C_{load} = 3\text{fF}$.

2.4. Analyze:➤ **2.4.1. Testcase:**

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT
Temp	25°C
V_{DD}	1 V
V_{in}	$V_{in} = 0.7\text{V}$
Current	$I_{D,NM1} = I_{D,NM2} = I_{D,NM3} = 50\mu\text{A}$

- However, in reality, we cannot choose V_{in} and V_{out} independently, and in **Experiment 1**, we also do not control the current. Experiment 2 is more practical: the requirements are
 - $V_{in} = 0.7\text{V}$ and $I_D = 50\mu\text{A}$
 - All devices in the circuit are in saturation
- Students are required to verify the device sizes in all three topologies to meet the specified requirements.

➤ **2.4.2. Analysis:**

- Students use the Calculator tool to perform measurements as shown in **Figure 7**.
- Write Calculator equations for all devices in **Figure 1**.
- Assume $V_{THN} = |V_{THP}| = 500\text{mV}$ for initial calculation.

✚ **Note:** Since only the fingers and multipliers are adjusted, the current does not need to be exactly 50 μA ; a tolerance of $\pm 5\%$ around 50 μA is acceptable.

✚ **Please capture the measured voltage and current in the Virtuoso tool to double-check that your results are valid.**

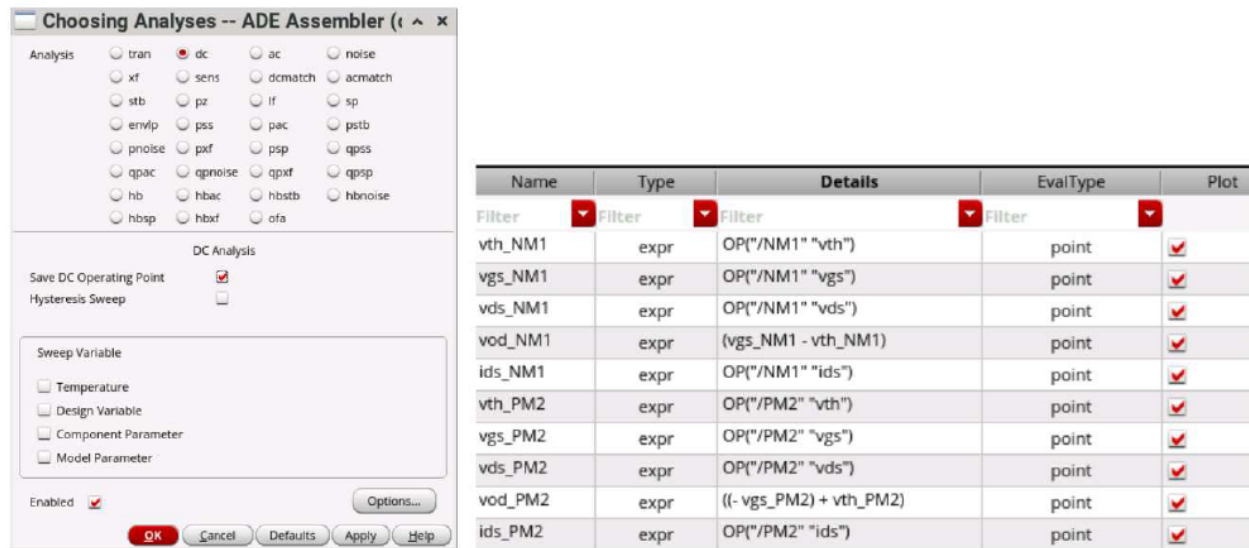


Figure 7. Setup for 2.4.2 section

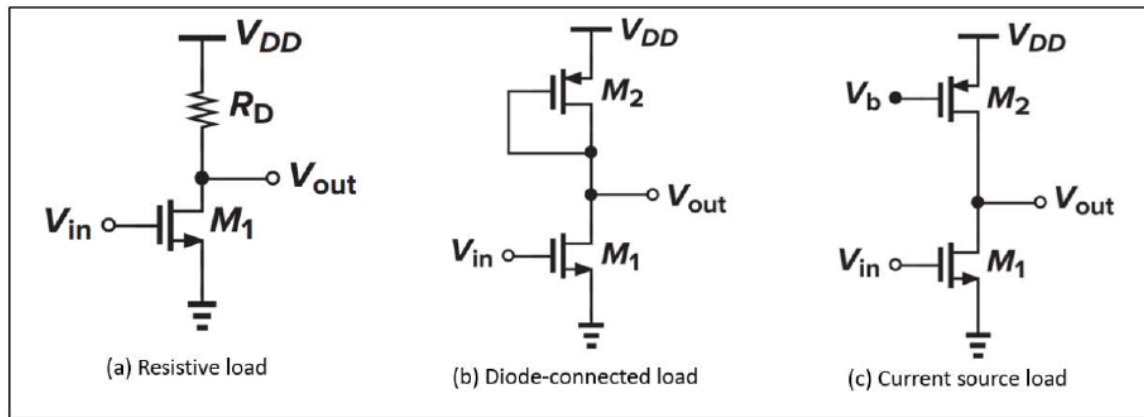


Figure 8. Common source stage topology

- Figure 8(a), verify the sizes of M1 and R_D that ensure $I_{D,M1} = 50\mu A$ with $V_{in} = 0.7V$. In addition, check V_{GS} , V_{DS} , V_{OV} of M1 to ensure it is biased in saturation.

Device name	Size	Voltage check
NM1 (M1)	$L_n = 45nm, W_n = 120nm,$ $fingers = \dots, multipliers = \dots$	$V_{TH} = \dots, V_{GS} = \dots, V_{DS} = \dots,$ $V_{OV} = \dots, I_D = \dots$ $\rightarrow$ M1 in ... region
PR1 (RD)	Segment Width = 1u, Segment Length = ... Segment = ..., Segment Connection = Series	

- Figure 8(b), verify the sizes of M1 and M2 that ensure $I_{D,M1} = 50\mu A$ with $V_{in} = 0.7V$. In addition, check V_{GS} , V_{DS} , V_{OV} of M1 and M2 to ensure it is biased in saturation.

Device name	Size	Voltage check
NM2 (M1)	$L_n = 45nm, W_n = 120nm,$ $fingers = ..., multipliers = ...$	$V_{TH} = ..., V_{GS} = ..., V_{DS} = ..., V_{OV} = ...,$ $I_D = ... \rightarrow M1 \text{ in } ... \text{ region}$
PM2 (M2)	$L_p = 45nm, W_p = 120nm,$ $fingers = ..., multipliers = ...$	$V_{TH} = ..., V_{SG} = ..., V_{SD} = ..., V_{OV} = ...,$ $I_D = ... \rightarrow M2 \text{ in } ... \text{ region}$

- Figure 8(c), verify the sizes of M1, M2 and voltage V_{bias} that ensure $I_{D,M1} = 50\mu A$ with $V_{in} = 0.7V$. In addition, check V_{GS}, V_{DS}, V_{OV} of M1 and M2 to ensure it is biased in saturation.

Device name	Size	Voltage check
NM3 (M1)	$L_n = 45nm, W_n = 120nm,$ $fingers = ..., multipliers = ...$	$V_{TH} = ..., V_{GS} = ..., V_{DS} = ..., V_{OV} = ...,$ $I_D = ... \rightarrow M1 \text{ in } ... \text{ region}$
PM3 (M2)	$L_p = 45nm, W_p = 120nm,$ $fingers = ..., multipliers = ...$	$V_{TH} = ..., V_{SG} = ..., V_{SD} = ..., V_{OV} = ...,$ $I_D = ... \rightarrow M2 \text{ in } ... \text{ region}$ $V_{bias} = ...$

➤ 2.4.3. Waveform:

- Using DC analysis, sweeping variable v_{in} from 0V to v_{dd} , step = 0.01V as shown in **Figure 2**. Plot IN-OUT characteristic curve as **Figure 4**.
- Method to calculate DC Gain, input max swing and output max swing as shown in **Figure 9**.

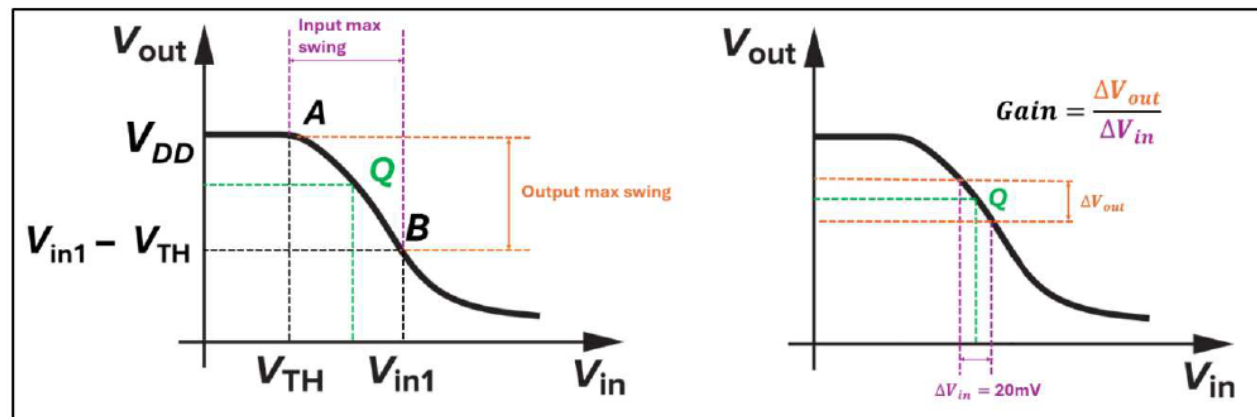


Figure 9. Method calculating Gain, Input max-swing and Output max-swing

- **Double check the Gain, Output swing and Input swing at $V_{in} = 0.7V$**

Topology	Optimize Q	Gain (dB) at Q with $\Delta V_{IN} = 20mV$	Max swing input	Max swing output
Resistive	$V_{IN} = 0.7V$ $V_{OUT} = \dots$			
Diode connected	$V_{IN} = 0.7V$ $V_{OUT} = \dots$			
Current source	$V_{IN} = 0.7V$ $V_{OUT} = \dots$			

- **Are the bias conditions in Experiment 2 the optimal Operating point Q for gain and swing?**

EXPERIMENT 3

3.1. Objective: Design amplifiers to meet given specifications.

3.2. Requirements: Simulate DC, AC and Transient analysis with ADE-Explorer environment.

3.3. Instruction: Assemble the circuit as shown in **Figure 10**, setting parameters for power supplies (using **vdc** source), with $L_n = L_p = 45\text{nm}$, $W_{n,\text{unit}} = W_{p,\text{unit}} = 120\text{nm}$, $v_{dd} = 1.0\text{V}$, $V_{in,DC} = 0.65\text{V}$, $C_{load} = 10\text{fF}$, $v_{in,AC} = 1.0\text{V}$, phase = 0.

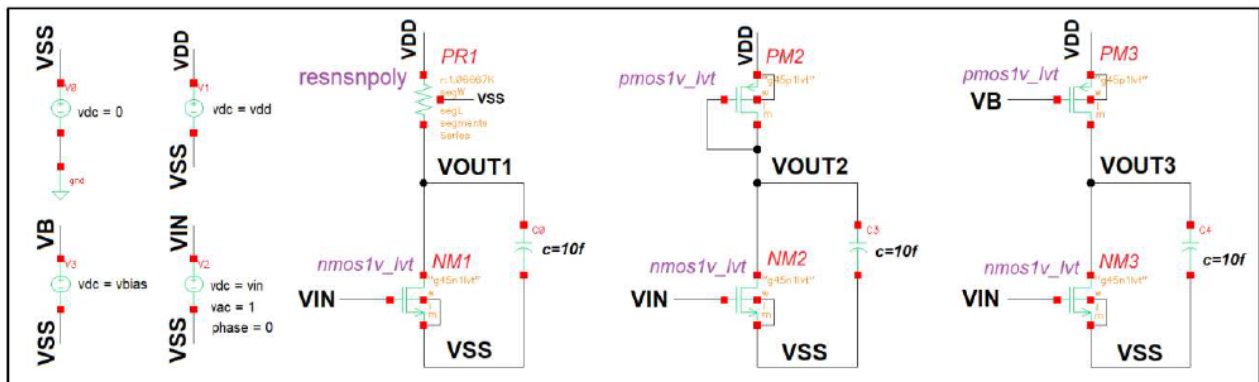


Figure 10. Common source stage schematic for DC and AC analysis

3.4. Analyze:

Topology	Resistive load	Diode-connected load	Current source load
PVT	TT – 1.0V – 25°C	TT – 1.0V – 25°C	TT – 1.0V – 25°C
Device	PMOS: <i>pmos1v_lvt</i> NMOS: <i>nmos1v_lvt</i>	PMOS : <i>pmos1v_lvt</i> NMOS : <i>nmos1v_lvt</i>	PMOS : <i>pmos1v_lvt</i> NMOS : <i>nmos1v_lvt</i>
I_D	70uA $\pm 10\%$	70uA $\pm 10\%$	70uA $\pm 10\%$
C_{load}	10fF	10fF	10fF
Input common mode	0.65V	0.65V	0.65V
Output common mode	...	...	...
Gain at DC	> 6dB	> 0dB	> 10dB
Output max swing (Headroom)	As much as possible	As much as possible	As much as possible
3-dB bandwidth	> 1GHz	> 1GHz	> 1GHz

✚ Step Procedure:

- **Step 1:** Determine the initial input and output voltage levels to ensure sufficient voltage headroom and achieve the desired gain under specified bias conditions.
- **Step 2:** Simulate and size device widths to meet the bias current condition $I_D = 70\mu A$, ensuring all devices operate in the saturation region at the desired V_{out} .
- **Step 3:** Perform DC analysis to verify bias conditions and small-signal gain.
- **Step 4:** Perform AC analysis to determine the -3dB bandwidth (3dB-BW) and unity-gain bandwidth (UGBW).
- **Step 5:** Perform transient simulation at the corresponding frequency to verify circuit response.

➤ 3.4.1. DC analysis

Device name	Size	Voltage check
NM1 (M1)	$L_n = 45nm, W_n = 120nm,$ $fingers = ..., multipliers = ...$	$V_{TH} = ..., V_{GS} = ..., V_{DS} = ...,$ $V_{OV} = ..., I_D = ...$ → NM1 in ... region
PR1 (RD)	Segment Width = 1u, Segment Length = ... Segment = ..., Segment Connection = ...	
NM2 (M1)	$L_n = 45nm, W_n = 120nm,$ $fingers = ..., multipliers = ...$	$V_{TH} = ..., V_{GS} = ..., V_{DS} = ...,$ $V_{OV} = ..., I_D = ...$ → NM2 in ... region
PM2 (M2)	$L_p = 45nm, W_p = 120nm,$ $fingers = ..., multipliers = ...$	$V_{TH} = ..., V_{SG} = ..., V_{SD} = ...,$ $V_{OV} = ..., I_D = ...$ → PM2 in ... region
NM3 (M1)	$L_n = 45nm, W_n = 120nm,$ $fingers = ..., multipliers = ...$	$V_{TH} = ..., V_{GS} = ..., V_{DS} = ...,$ $V_{OV} = ..., I_D = ...$ → NM3 in ... region
PM3 (M2)	$L_p = 45nm, W_p = 120nm,$ $fingers = ..., multipliers = ...$	$V_{TH} = ..., V_{SG} = ..., V_{SD} = ...,$ $V_{OV} = ..., I_D = ..., V_{bias} = ...$ → PM3 in ... region

- Verify the sizing of NM1, PR1, NM2, PM2, NM3, and PM3 to confirm compliance with the design specifications, and ensure that all devices operate in the saturation region.

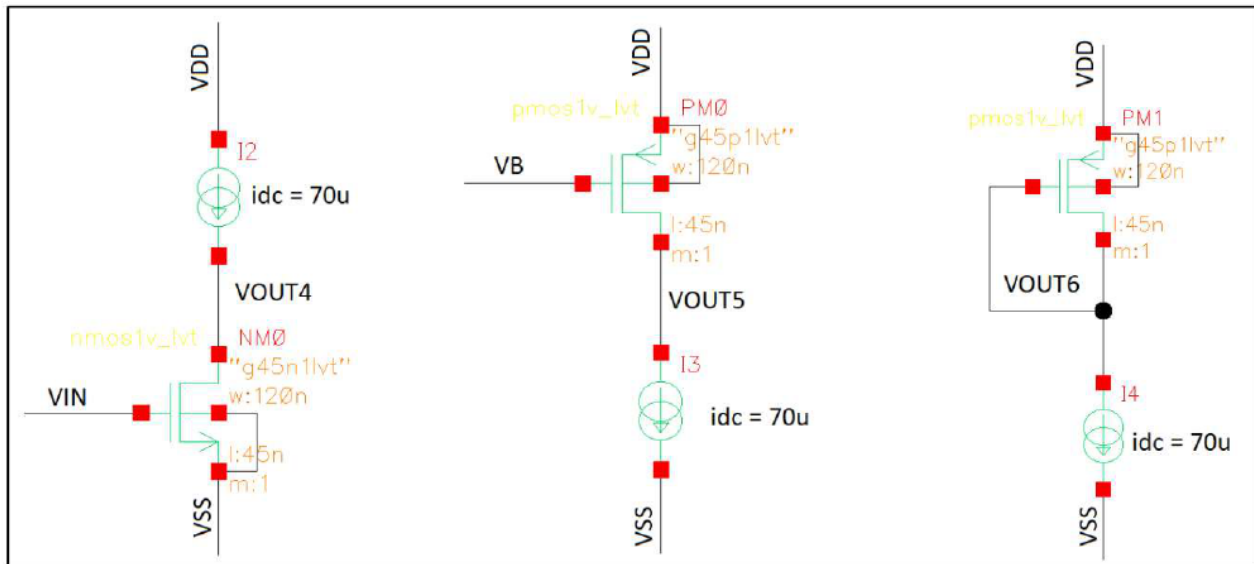


Figure 11. Method calculating Gain, Input max-swing and Output max-swing

- Students may use an ideal current source to determine the MOS device sizing, with V_{IN} , V_{OUT} and V_B properly configured to meet the specified conditions, as shown in **Figure 11**. **Note that MOS device sizing can only be modified by adjusting the multipliers and number of fingers, while the width and length must remain unchanged.**
- Plot IN-OUT characteristic curve using DC analysis. **Calculate Gain, Output max swing (Headroom) and Input max swing** as shown in **Figure 12**.

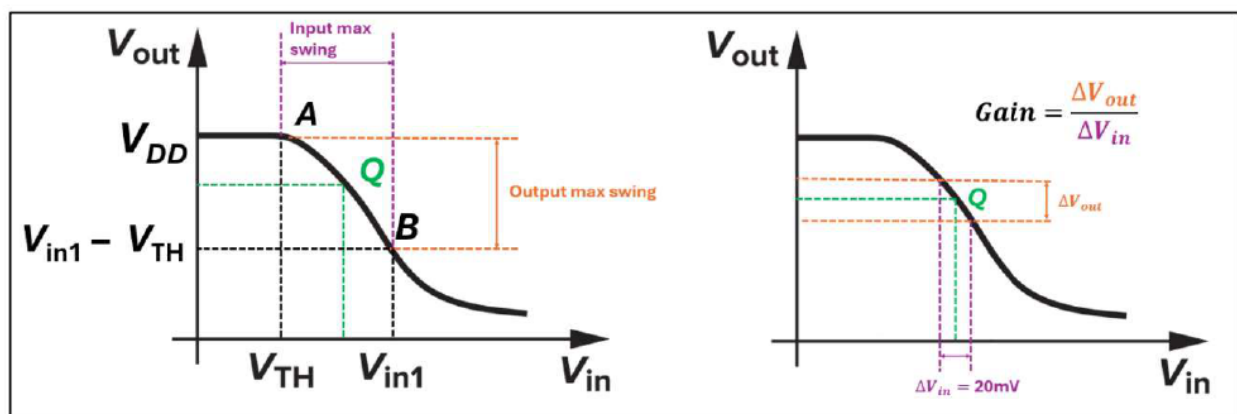


Figure 12. Method calculating Gain, Input max-swing and Output max-swing

Topology	Optimize Q	Gain (dB) at Q with $\Delta V_{IN} = 20mV$	Max swing input	Max swing output
Resistive	$V_{IN} = 0.65V$, $V_{OUT} = \dots$			
Diode connected	$V_{IN} = 0.65V$, $V_{OUT} = \dots$			
Current source	$V_{IN} = 0.65V$, $V_{OUT} = \dots$			

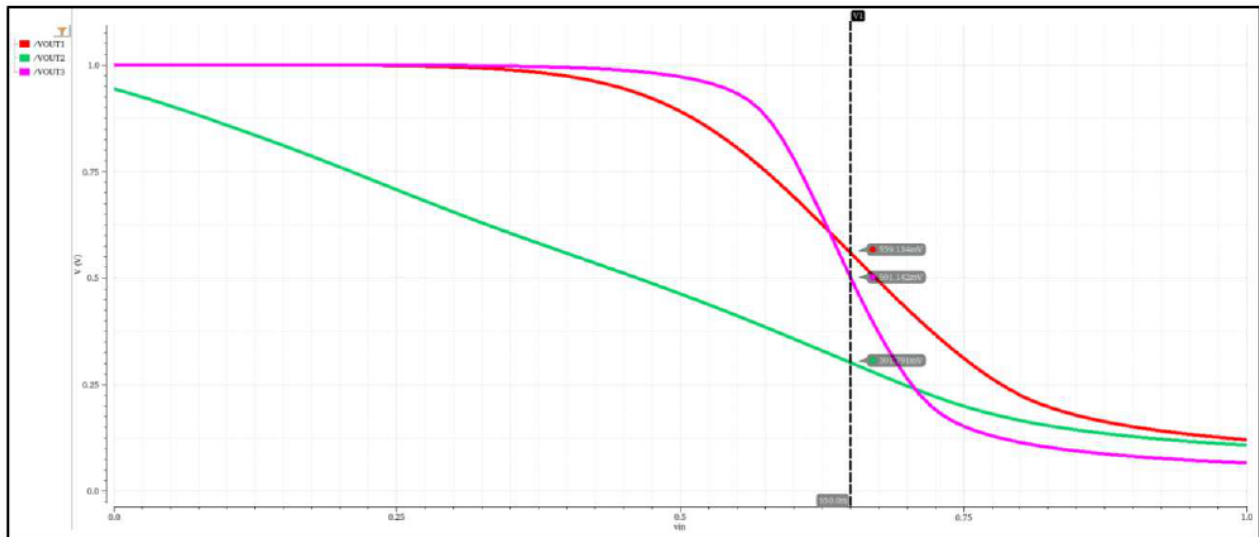


Figure 13. DC analysis result

- Figure 14 shows that all devices are biased in the saturation region.

Parameter			TTNN
gpdtk045.scs			tt
temperature			25
vdd			1
Output	Min	Max	TTNN
Filter	Filter	Filter	Filter
vds_NM1	559.1m	559.1m	559.1m
vod_NM1	159.7m	159.7m	159.7m
ids_NM1	74.4u	74.4u	74.4u
vds_NM2	301.8m	301.8m	301.8m
vod_NM2	154.2m	154.2m	154.2m
vds_PM2	-698.2m	-698.2m	-698.2m
vod_PM2	216m	216m	216m
ids_PM2	-69.66u	-69.66u	-69.66u
vds_NM3	501.1m	501.1m	501.1m
vod_NM3	158.9m	158.9m	158.9m
vds_PM3	-498.9m	-498.9m	-498.9m
vod_PM3	17.13m	17.13m	17.13m
ids_PM3	-71.54u	-71.54u	-71.54u

Figure 14. DC analysis result

➤ 3.4.2. AC analysis

DC voltage	vin V
AC magnitude	1 V
AC phase	0

Figure 15. Setup AC voltage for AC analysis

- Set up VIN voltage as shown in **Figure 15**, AC magnitude = 1.0V and AC phase = 0. **Draw frequency response with ac analysis, frequency varies from 1Hz to 10GHz as Figure 16.**

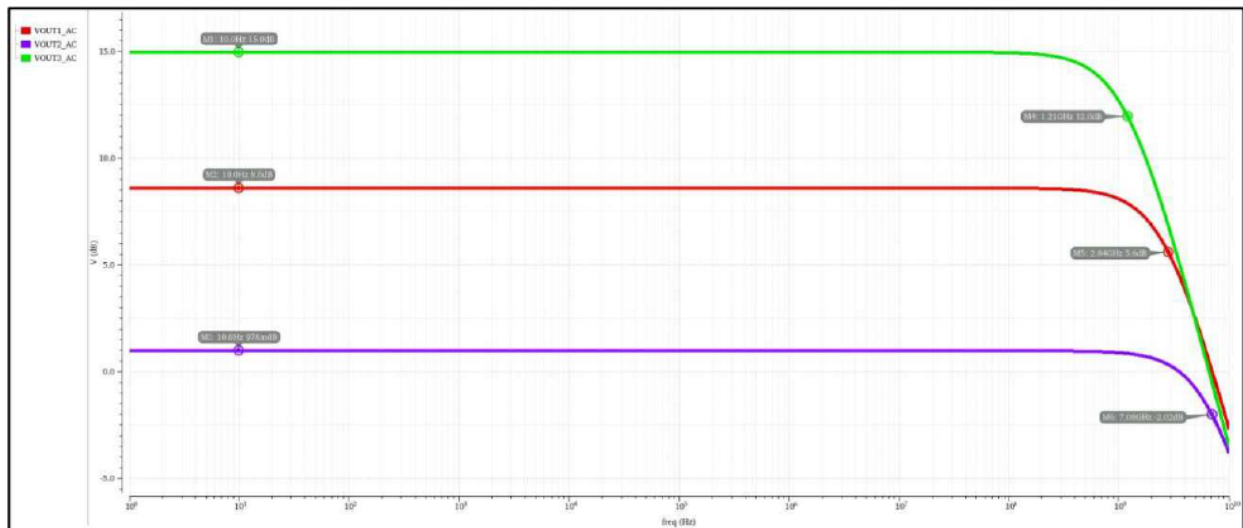


Figure 16. AC response

- Determine the DC gain and 3dB Bandwidth.**

Parameter	TTNN		
gpd045.scs	tt		
temperature	25		
vdd	1		
Output	Min	Max	TTNN
Filter	Filter	Filter	Filter
vds_NM1	559.1m	559.1m	559.1m
vod_NM1	159.7m	159.7m	159.7m
ids_NM1	74.4u	74.4u	74.4u
vds_NM2	301.8m	301.8m	301.8m
vod_NM2	154.2m	154.2m	154.2m
vds_PM2	-698.2m	-698.2m	-698.2m
vod_PM2	216m	216m	216m
ids_PM2	-69.66u	-69.66u	-69.66u
vds_NM3	501.1m	501.1m	501.1m
vod_NM3	158.9m	158.9m	158.9m
vds_PM3	-498.9m	-498.9m	-498.9m
vod_PM3	17.13m	17.13m	17.13m
ids_PM3	-71.54u	-71.54u	-71.54u
VOUT1_GAIN	8.601	8.601	8.601
VOUT2_GAIN	978.4m	978.4m	978.4m
VOUT3_GAIN	14.96	14.96	14.96
VOUT1_BW_3db	2.838G	2.838G	2.838G
VOUT2_BW_3db	7.08G	7.08G	7.08G
VOUT3_BW_3db	1.207G	1.207G	1.207G

Figure 17. DC and AC analysis result

➤ 3.4.3. Transient analysis

- Replace the V2 source from a **vdc** type to a **vsin** type. Set the **offset voltage equal to $V_{in} = 0.65V$, frequency = 1.0GHz and amplitude = 10mV** as shown in **Figure 18**.

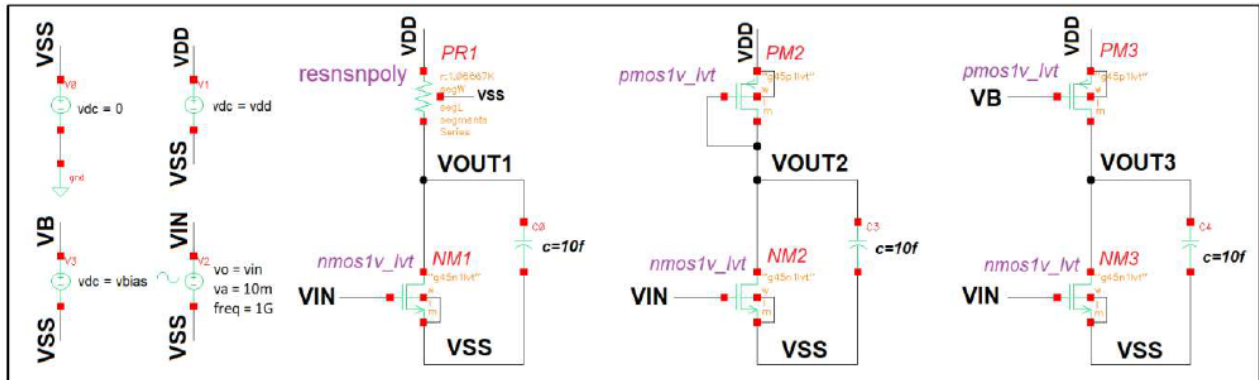


Figure 18. Common source stage schematic for TRAN analysis

- Perform transient analysis from 0 s to 100ns and plot the input and output waveforms.
- Calculate the gain from the input and output signal amplitudes, explain the observed waveform behavior, and compare the results with those from DC and AC analyses.**

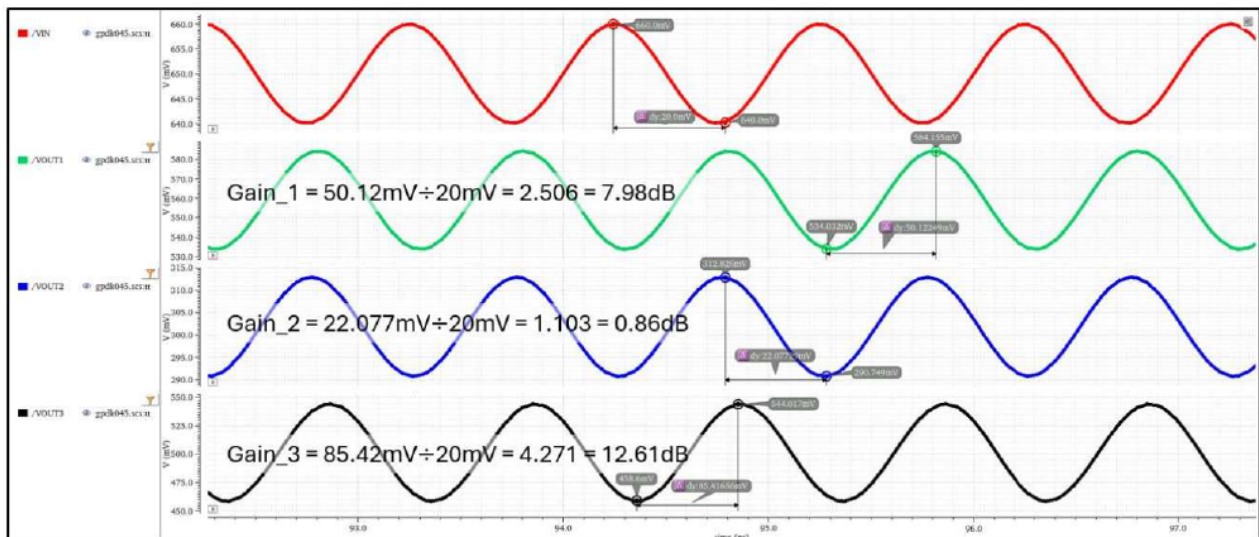


Figure 19. TRAN analysis result

❖ **Wrap-up Questions – Lab 4: Study of Common-Source Single-Stage Amplifiers**Complete **Table 1** and **Table 2****Table 1**

CS Stage	A_v	Z_{in}	Z_{out}	Input swing	Headroom	Linearity
Resistive load	$-g_m(R_D r_o)$	∞	$R_D r_o$	$V_{TH} \rightarrow V_{out} + V_{TH}$	$V_{in} - V_{TH} \rightarrow V_{DD}$	g_m depends on V_{in}
Diode Connected load						
Current Source load						
Active load						
Triode load						
Source degeneration						

Table 2

Topology	Advantages	Disadvantages
Resistive load	• Large bandwidth • Large headroom	• Resistor need large area • Non-linearity
Diode connected	•	•
Current-source	•	•
Active load	•	•
Triode load	•	•
Source degeneration	•	•

